Differential manifolds.

I. Basic concepts

1 Smooth manifolds & functions, maps on smooth manifolds

Def I.1.1 (Topological manifold) M is a topology space is a manifold iff.

(i) 本 M is T_2, (ii) M is secondly-countable, (iii) $\forall x \in M$. $\exists$ Nbh(x) s.t. Nbh(x) is homeomorphism to $\mathbb{R}^n$, $\varphi(x)$ open 且 $\exists$ Ball $B^\circ \subseteq U$, $\varphi(x) \in B^\circ$

Rmk. Nbh(x) 在 $\mathbb{R}^n$ can be replaced equivalently by $U \subseteq \mathbb{R}^n$ open. ($x \in U_x \subseteq M$, $\varphi(x) \in U$)

Def I.1.2. (Differential manifold) M is a n- topological manifold. called C^r- differential manifold. iff $\exists$ $M \subseteq \bigcup_{\alpha \in \Gamma} U_\alpha$, $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$. continuous.

s.t. (1) φ_α is a homeomorphism

(2) $U_\alpha \cap U_\beta \neq \phi$, we have, $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ is C^r- map.

Then we call M is a C^r- manifold.

When $r = \infty$, we call M is a smooth - manifold.

Rmk. 1) $(U_\alpha, \varphi_\alpha)$ is called coordinate chart.

2) U_α is the coordinate domain, φ_α is the coordinate map.

Def I.1.3. M is a C^r- manifold, f is called a smooth function : $M \to \mathbb{R}^k$:

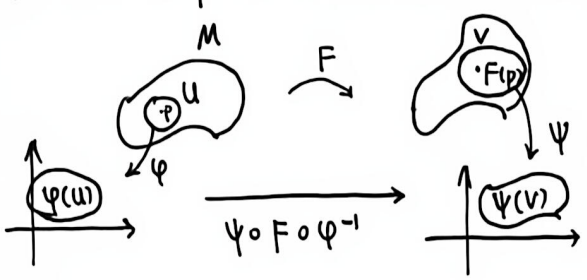
If 对 $\forall p \in M$. $\exists$ smooth chart (U, φ) where $p \in U$. then $f \circ \varphi^{-1}$ is smooth on the set $\hat{U} = \varphi(U) \subseteq \mathbb{R}^n$.

Def I.1.4. Let M, N be smooth manifolds, $F : M \to N$ is a smooth map iff

$\forall p \in M$. $\exists$ smooth chart (U, φ) and (V, ψ) containing $F(p)$, $F(U) \subseteq V$

s.t. $\psi \circ F \circ \varphi^{-1}$ is smooth from $\varphi(U)$ to $\psi(V)$.

Rmk. It's easy to check that the above are well-defined.



Def I.1.5 $f: M \to N$ be the smooth map. If f is a bijection and $f^{-1}: N \to M$ is smooth

Then we call f is a diffeomorphism from M to N.

2. Manifolds with boundary. (Differential)

Def I.2.1. M is an n-dim topological manifold with boundary iff

(1) M is Hausdorff;

(2) M has a second-countable basis

(3) $\forall x \in M.\ \exists U_x \subseteq M$ open s.t. U_x is homeomorphic either $U \subseteq \mathbb{R}^n$ open or $V \subseteq \mathbb{H}^n$ open, where $\mathbb{H}^n := \{(x^1, \dots, x^n) \in \mathbb{R}^n \mid x^n \geq 0\}$

Rmk. (1) (U, φ) is an interior chart if $\varphi(U)$ is open in $\mathbb{R}^n$ (*)

(U, φ) is a boundary chart if $\varphi(U)$ is open in $\mathbb{H}^n$ with $\varphi(U) \cap \partial\mathbb{H}^n \neq \phi$ (**)

(2) $p \in M$ is an interior point if $\exists (U, \varphi)$ (*) s.t. $p \in U$

$p \in M$ is a boundary point if $\exists (U, \varphi)$ (**) s.t. $p \in U$

Thm I.2.2. M^n is a manifold with boundary. Then $\forall p \in M^n$, p is a boundary point or interior point, but not both.

proof: From the definition, it suffices to show that p can not be both boundary or interior.

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Def I.2.3. M^n is a manifold with boundary. Then M^n is a smooth manifold if and only if: $\exists\ M^n \subseteq \bigcup_{\alpha\in\Gamma} U_\alpha$, $\varphi_\alpha : U_\alpha \to \mathbb{R}^n$ ($\mathbb{H}^n$) continuous s.t. (1) φ_α homeomorphism

(2) If $U_\alpha \cap U_\beta \neq \phi$, we have: $\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \to \varphi_\beta(U_\alpha \cap U_\beta)$ smooth.

3. Topology properties of manifolds

Prop. I.3.1. Every topological manifold M has a countable basis of precompact coordinate balls.

proof: Case 1. M has a global coordinate map. say (M, φ).

$\varphi: M \to \mathbb{R}^n$ Let $\mathcal{B} = \{B \mid B = B_r(x),\ x \in \mathbb{Q}^n,\ r \in \mathbb{Q}^+\}$.

Then $\mathbb{R}^n \subseteq \bigcup_{B\in\mathcal{B}} B$. , $|\mathcal{B}|$ is countable

Then $\mathcal{B}' = \{\varphi^{-1}(B) \mid B \in \mathcal{B}\}$. is a countable basis of precompact coordinate balls.

General case: Since M is second-countable. $\Rightarrow \exists$ countable-subcover of M: $\{(U_i, \varphi_i)\}$

Each U_i has a countable basis of precompact coordinate balls in U_i

$\Rightarrow \forall\ V \subseteq U_i$ precompact

Since M is T_2 $\Rightarrow \overline{V}$ compact in M. □.

Prop. I.3.2. Let M be a topological manifold.

(a). M is l.p.c.

(b) M is connected iff M is p.c.

(c) The components of M are the same as the path components

(d) M has countably many components, each of which is open in M and a connected topological manifold.

proof: (skipped) □.

Prop. I.3.3. Manifolds is locally compact. precompact.

proof: $\{(U_n, \varphi_n)\}_n$ is the countable coordinate balls.

$\Rightarrow \forall\ x \in M$, $\exists\ U_k$. s.t. $x \subseteq U \subseteq U_k \subseteq \overline{U_k}$. $\overline{U_k}$ is compact □.

($\forall\ x \in U$ open.)

Prop. I.3.4. (Manifolds are paracompact 仿紧.) Given a topological manifold M, an open cover $\mathcal{X}$ of M, and any basis $\mathcal{B}$, $\exists$ countable, locally finite open refinement of $\mathcal{X}$ consisting of elements of $\mathcal{B}$

Proof: M is locally pocompact & Hausdorff. $\Rightarrow \exists\ K_1 \subseteq K_2 \subseteq \dots \subseteq K_n \subseteq \dots$ compact.

$M = \bigcup_{n\geq 1} K_n$. let $V_j = K_{j+1} \setminus \mathrm{Int}\, K_j$. , $W_j = \mathrm{Int}\, K_{j+2} \setminus K_{j-1}$.

Then $V_j \subseteq W_j$, V_j compact.

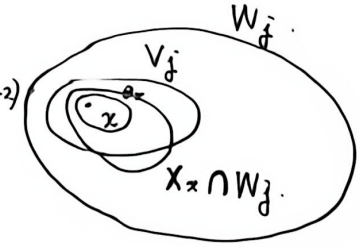
$\forall\ x \in V_j$. $\exists\ X_x \in \mathcal{X}$, $B_x \in \mathcal{B}$. s.t. $x \in B_x \subseteq X_x \cap W_j$.

$\Rightarrow \exists$ finite-subcover V_j. say $\{B_{ij}\}_{i\in I_j}$ $\Rightarrow \bigcup_{j\geq 1} B_{ij}$ is a countable subcover

$\Rightarrow \bigcup_{i,j} (\chi_{x_{i,j}} \cap W_j)$ is a countable cover of M.

By the constraction, $W_j \cap W_{j'} = \phi$ ($j' < j-2$ or $j' > j+2$)

□



Examples of smooth manifolds.